Additional file 4: Track portability

This supplement concerns the same 84 sites and seven genes under a second AlphaGenome contact track. It is not a new cohort or independent model. The protocol was locally frozen before acquisition, then native-only predictions were locked before centered evaluation.

Technical transfer: PASS. Operational benchmark: PASS. The operational threshold was at least 17 additional leaders and gains in at least four genes, with no lost simple-bound decisions, complete data and no validation failures. It is not a powered hypothesis test.

Gene-level primary results (tie tolerance 1e-6)

Gene	Sites	Simple leaders	Strengthened leaders	Additional	Native credit	Centered credit
ASS1	12	3	10	7	0.250000	0.083333
BAP1	12	0	11	11	0.333333	0.750000
BRCA2	12	1	8	7	0.305556	0.416667
HFE	12	0	4	4	0.347222	0.375000
RB1	12	2	2	0	0.444444	0.416667
SCN5A	12	0	6	6	0.347222	0.416667
TSC1	12	1	8	7	0.319444	0.333333

Fixed tolerance sensitivities

Tie tolerance	Simple leaders	Strengthened leaders	Additional	Native ties	Centered ties
0.0	7	50	43	75	0
1e-05	7	42	35	75	12
1e-06	7	49	42	75	1

Fixed baseline comparisons

Score	IMR-90 selected credit	GM12878 selected credit	IMR-90 agreement with centered top set
absolute mean change	0.410714	0.315476	58/84
centered	0.398810	0.321429	84/84
centered rms	0.517857	0.460317	36/84
lifetime count absolute change	0.349206	0.434524	33/84
max absolute	0.341270	0.357143	10/84
maximum persistence absolute change	0.353175	0.327381	16/84
native	0.335317	0.359127	9/84
native minus absolute mean change	0.244048	0.380952	16/84
native plus absolute mean change	0.410714	0.321429	64/84
native rms	0.500000	0.444444	39/84
total persistence absolute change	0.464286	0.333333	21/84

Deleting one gene leaves 31–42 additional leaders among 72 sites. The corresponding centered-minus-native credit ranges from +0.004630 to +0.101852. These are concentration diagnostics, not independent-sample uncertainty intervals.

Provenance, controls and precision

Frozen source hashes and timestamps, all acquisition attempts and the exact per-site selections are retained in the experiment bundle. The two central SHA256 records are:

```
Protocol:
56c5ae0650a02f05384e1d1cbc19f894d957d155f196a6db5432f9f36e10aa43
Native-prediction lock:
cc6ddc14d531f8e95974dbb69ae1b8dc3b7fdf73ed0f4201279836e1b52cef40
```

Both track channels were selected by exact metadata name. All 252 requests succeeded on their first attempt. Every repeated REF agreed, every contemporary GM12878 map equaled its historical counterpart, and every paired IMR-90 REF/ALT map differed from GM12878. All 132,120,576 delivered float32 values per track, including repeated reference arrays, were exactly bfloat16-representable. Equal outputs at these inputs do not establish immutable weights or biological measurement precision.

Nine phased-engine controls tested scientific equivalence, missing/changed locks, changed protocol, nonsaturation and wrong-leader detection. Twelve summary boundary controls tested the operational thresholds and incomplete data. The full replay checked 3696 scientific comparisons; 3696 were exactly equal. The independent per-site audit reports status PASS and maximum Hausdorff difference 0.

Bounds apply to the stored floating-point diagrams with a practical 1e-12 allowance, not a rigorous arithmetic enclosure. The matching solver uses zero approximation parameter. Added PH input rounding, observed grid structure and identical repeated outputs are separately described; none recovers latent model uncertainty. No runtime advantage, clinical accuracy improvement or experimental chromatin mechanism is established. No new clinical testing family was introduced.

The full protocol, machine-readable summary, independent audit, replay comparison and preserved execution correction accompany this supplement. Original discovery and replication evidence remains separately reported.

Post-result direct path diagnostics

After the primary results were known, a separate local protocol specified direct evaluations of all three nonsaturating comparisons on the grid lambda = 0, 0.1, 0.25, 0.5, 0.75, 0.9 and 1. The diagnostic protocol and code were hashed before these additional queries. Selection was explicitly outcome-dependent and included all exceptions, with no change to the original cohort, primary endpoint or saved predictions.

RB1 site (GRCh38), alternate index	Sampled behavior	Maximum departure from endpoint chord
chr13:48303721:G, 2	Direction changes on sampled grid	0.000903
chr13:48307406:G, 1	Direction changes on sampled grid	0.00129
chr13:48465370:T, 0	Flat on sampled grid	1.08e-16

Alternate indices are zero-based in the saved alphabetical non-reference allele order. All 21 direct distances satisfied the two-endpoint envelopes, and endpoint recomputations agreed within 1e-12. Both simple and strengthened bounds had already abstained at all three sites at all fixed tolerances. This is conservative behavior shared by both bounds; it is not the witness-specific incremental advantage.

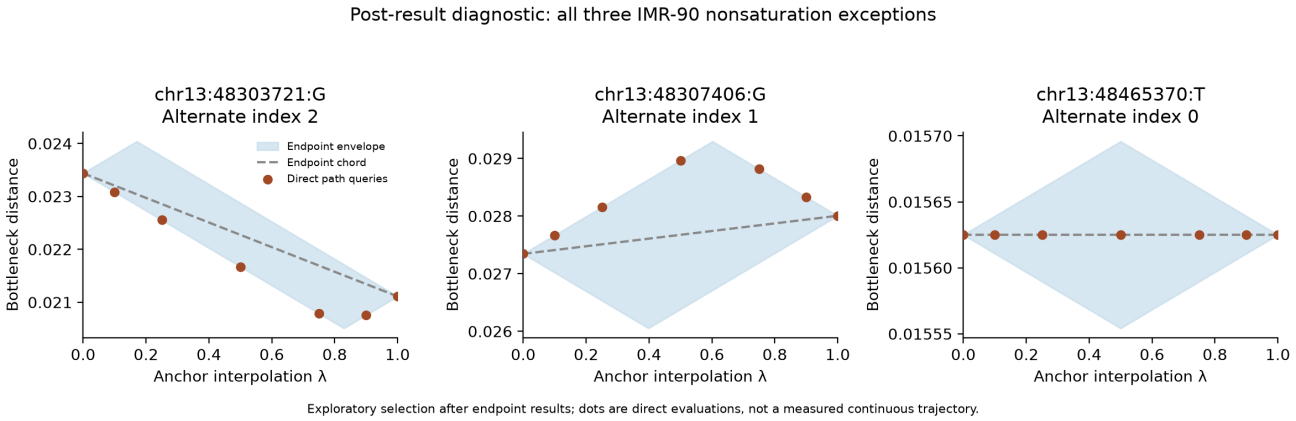


Figure S4.1. Post-result exploratory calculations for all three RB1 saturation exceptions. Points are direct matching computations at the seven specified lambda values and do not certify intervening behavior. Endpoint envelopes use their analytical kink vertices. Two sampled paths change direction, while one is effectively flat. Dashed chords are shown as a comparison, not an assumed interpolation. No exact turning location or matching identity is inferred.